

11. Device Management & File System Management

Computer Science Fundamentals | Module 3: Storage & I/O Subsystem | CS_Fundamental-00

Interview Memory Hook: Hazaaron alag-alag companies ke hardware ke liye developer ko alag code na likhna pade, isliye OS **Device Drivers** deta hai. Aur storage ke raw blocks ko sundar folders mein **File System** badalta hai!

11.1 Device Management (I/O Subsystem)

- **Device Drivers:** Hardware-specific translators jo generic commands (read/write) ko hardware signals mein badalte hain.
- **Uniform Interface:** Application code ko hardware details se decouple karta hai.
- **Activities:** Tracking connected devices, designating I/O controllers, scheduling device access, safe allocation & deallocation.

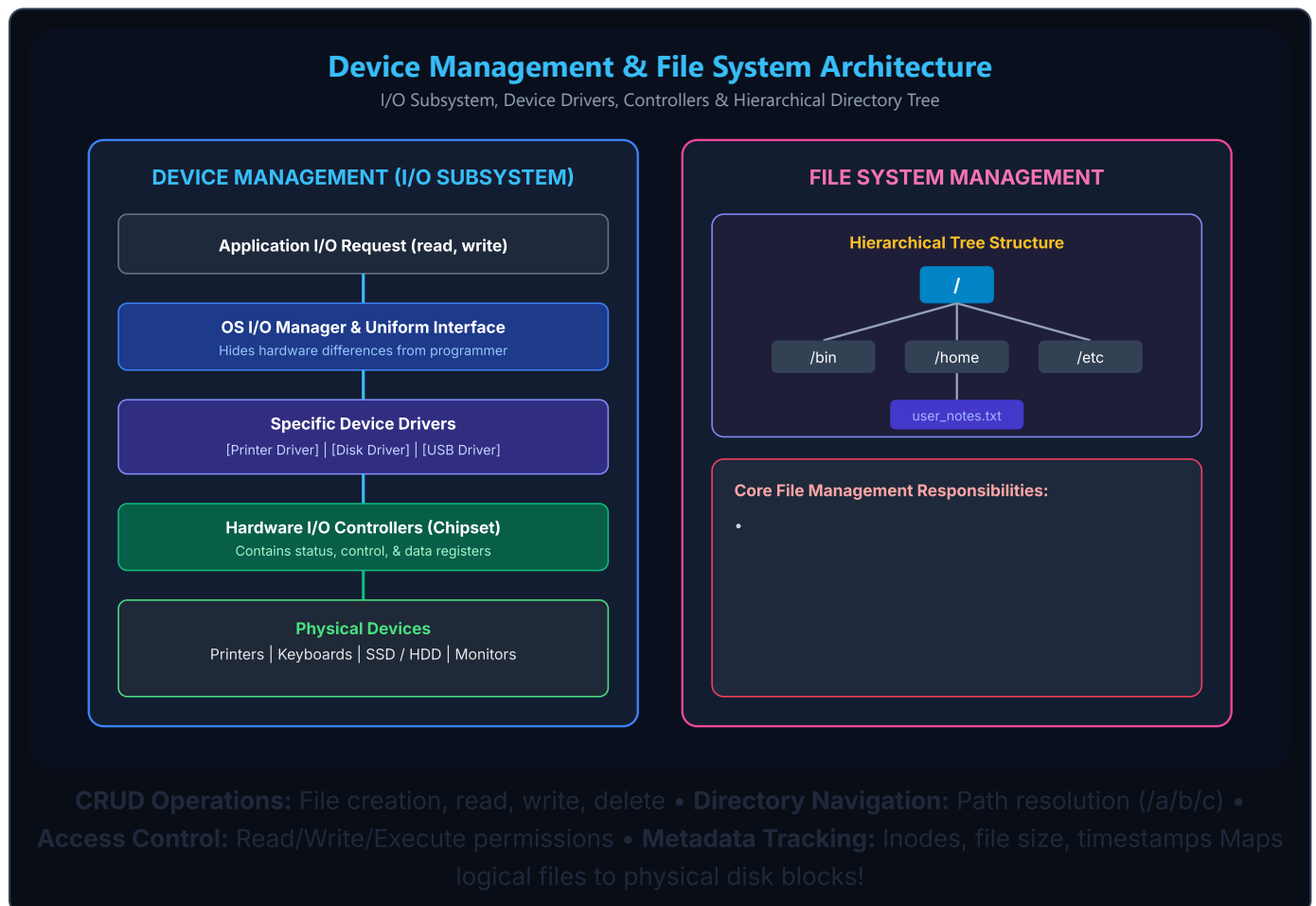


Fig 11.1: Device Driver Subsystem & Hierarchical File System Tree

11.2 File System Management

- **Logical Abstraction:** Physical disk blocks ko logical Files & Directories mein map karna.

- **Hierarchical Tree:** Root directory /, subdirectories, path resolution.
- **File Operations:** Create, Read, Write/Update, Delete, Open, Close.
- **Metadata & Inodes:** File size, permissions (rwx), block pointers, timestamps.

11.3 Step-by-Step Thought Process: File Read Journey (open("notes.txt", "r"))

Execution Trace:

1. **System Call:** User mode app invokes `sys_open()`.
2. **Path Resolution:** Directory traversal: `/` → `home` → `shivam` → `notes.txt`.
3. **Permission & Inode Check:** OS verifies read permission bit.
4. **Block Translation:** Inode maps logical offset to physical disk blocks.
5. **Device Driver I/O:** Disk controller reads blocks into RAM buffer.
6. **File Descriptor Return:** OS returns fd index to user application.